Universidade do Minho

Ano Letivo: 2025/26

Bases de Dados NoSQL

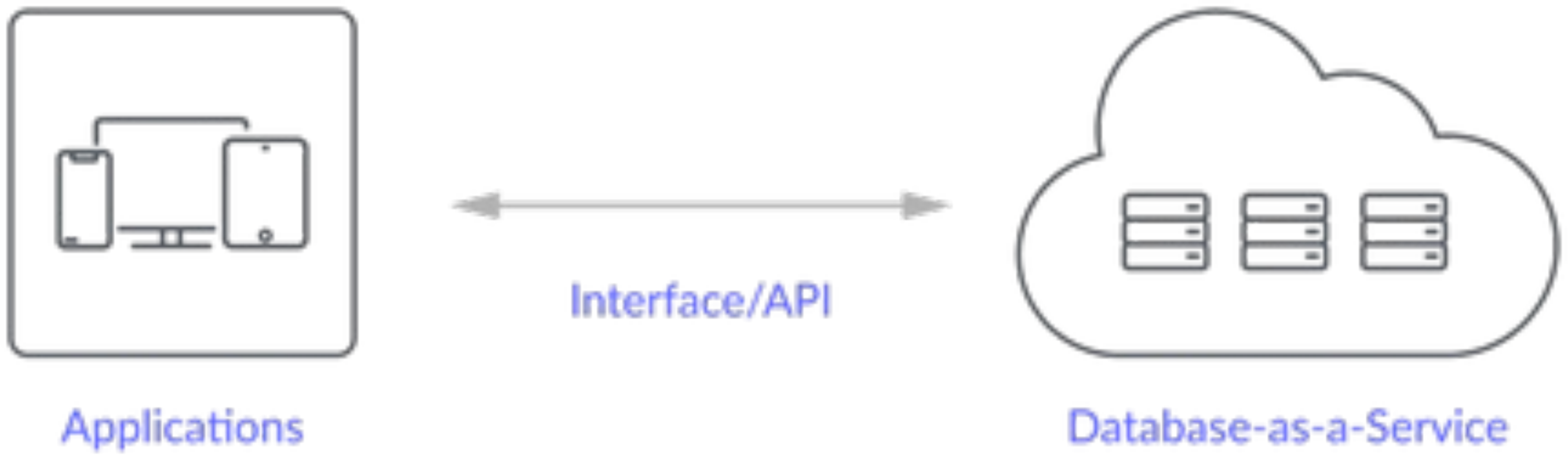
T05 – Database as a Service (DBaaS)

02

Database as a Service (DBaaS)

What is it?

Database-as-a-service (DBaaS) is a cloud computing service. As a hosted/managed service, users don't have to worry about setting up hardware or installing software. Everything related to managing the database is handled by the service provider.



Database hosting options are available for all database types, including NoSQL, MySQL, and PostgreSQL.

Database as a Service (DBaaS)

➜ Advantages

- **Reduced Management Overhead:** Eliminates the need for organizations to handle physical hardware setups or complex database software configurations.
- **Scalability:** Easily scales database resources up or down based on demand, without needing significant upfront investment in physical hardware.
- **Cost Efficiency:** Typically operates on a pay-as-you-go model, which can significantly reduce upfront and operational costs

Database as a Service (DBaaS)

➡ Challenges

- **Data Security:** Security concerns arise, particularly in multi-tenant environments where databases are hosted on shared infrastructure.
- **Data Sovereignty:** Compliance with data residency and sovereignty laws can be complex, depending on where data is physically stored.
- **Vendor Lock-in:** Dependency on a particular cloud provider's tools and services can limit flexibility and potentially increase future migration costs.

Database as a Service (DBaaS)

Key Features of DBaaS

- **Automated Backups and Recovery:** Ensures that data is regularly backed up and can be quickly restored in case of data loss.
- **Performance Monitoring:** Tools and dashboards provided by the service offer insights into performance metrics, helping optimize database operations.
- **Security Measures:** Includes built-in security features such as encryption, access control, and audit logs to protect sensitive data.

Database as a Service (DBaaS)

Use Cases

Web and Mobile Applications

DBaaS is ideal for supporting the backend of web and mobile applications, offering easy scalability as user numbers grow.

E-commerce Platforms

Supports high transaction volumes and fluctuating demands typical in e-commerce scenarios.

Big Data Analytics

Cloud databases can efficiently handle large data sets and complex queries, providing the computational power needed for big data analytics.

Database as a Service (DBaaS)

Examples of DBaaS Providers

Cloud SQL

ORACLE CLOUD

MongoDB Atlas

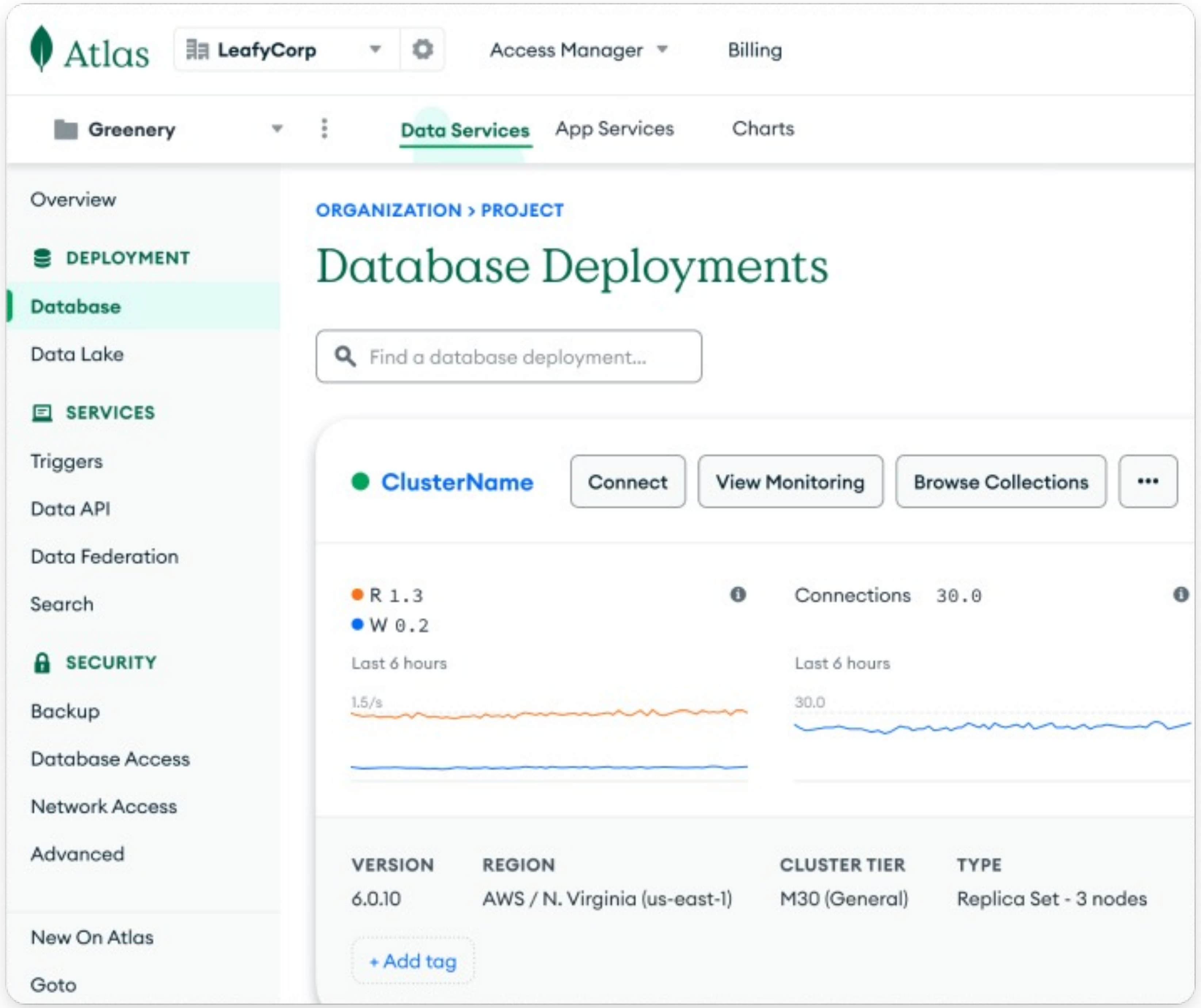
Amazon DynamoDB

Database as a Service (DBaaS)

➡ MongoDB Atlas

- MongoDB Atlas is a multi-cloud database service by the same people that build MongoDB.
- Live Migration Service
- Managed Cloud Database
- User focuses on development
  - Deploy, Manage, Restore with ease

MongoDB provides a free-tier (M0 Instances)



Database as a Service (DBaaS)

→ MongoDB Atlas

Cluster Storage

- Distributed workload
- Replication
- Increase redundancy and availability
- Sharded Cluster Components
- MongoDB free-tier defaults to a 3 node, replica set
  - Replica sets: MongoDB servers
  - Redundancy and Data Availability

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Advantages of MongoDB Atlas

Atlas Security

- Automated Security Features
- Authentication providers (Google Oauth, Facebook Oauth, API key, CLI Authentication through MongoDB credentials)
- Access control
- Encryption

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Advantages of MongoDB Atlas

Atlas Monitoring: Alerts, Monitoring, Live Analysis

- Web GUI Interface
  - Monitor a cluster’s performance
  - Performance metrics:
  - Alerts
- MongoDB compass
  - Visually explore your data
  - View and optimize your query performance

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➡ Advantages of MongoDB Atlas

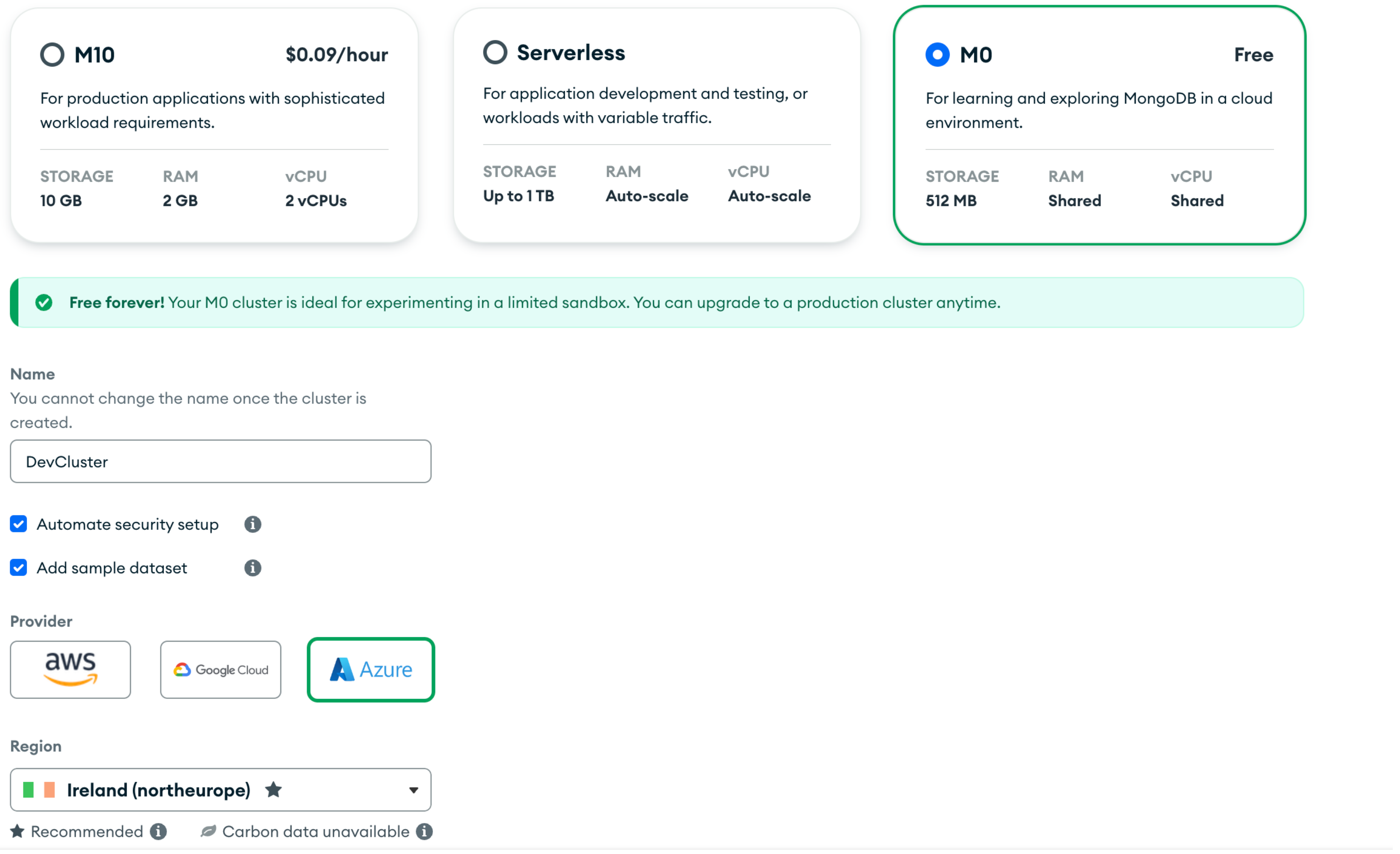
Atlas Backups

- Built-in Replication
- Backups and Point-In-Time Recovery
- Fine-Grained Monitoring

Database as a Service (DBaaS)

➜ Start using MongoDB Atlas

1. Sign up for MongoDB Atlas for free.
2. MongoDB Atlas provides a self-explanatory wizard to create the Organization, Project, and admin user components for the initial setup.
3. Once the initial setup is complete, it is time to build a free MongoDB cluster in a few easy steps. Start by clicking on the "Build Cluster" button and select the Free option.
4. Then pick the Cloud Provider and Region. In this example, I am going to pick Azure Ireland and update the Cluster Name to be "DevCluster." Then click "Create Cluster."



Database as a Service (DBaaS)

Start using MongoDB Atlas

Connect to Your Cluster

Step 1: Get the Connection String

1. Go to **Database** → **Connect**.
2. Choose "**Connect Your Application**".
3. Copy the provided **MongoDB URI** (looks like mongodb+srv://user:password@cluster.mongodb.net/myDatabase).

Step 2: Connect via MongoDB Compass (GUI)

1. Open Compass → Paste the **MongoDB URI** → Click **Connect**.

You can also access via Python.

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